Two-Maturity FTPL with Long-End Captive Demand: Derivation

1 Primitives

Time $t = 0, 1, 2, \ldots$. Single non-storable consumption good. All variables real except where indicated. Nominal: price level P_t, bond stocks B_t^S, B_t^L, bond prices Q_t^S, Q_t^L. Real bond stocks: $b_t^S \equiv B_t^S/P_t$, $b_t^L \equiv B_t^L/P_t$. Inflation: $\pi_t \equiv P_t/P_{t-1}$.

Parameters: $\beta \in (0,1)$, $\sigma > 0$, $\delta \in [0,1)$, $\bar{\lambda} > 0$.

Exogenous sequences: $\{y_t, g_t, \tau_t, \lambda_t^L, \chi_t, b_t\}_{t\geq 0}$ with $b_t \equiv b_t^S + b_t^L$. Exogenous structural constant $V^{\text{liab}} > 0$.

Define $s_t \equiv \tau_t - g_t$.

2 Household Problem

2.1 Statement

Maximize

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \tag{1}$$

subject to, for all $t \geq 0$,

$$c_t + Q_t^S b_t^S + \Phi_t = y_t - \tau_t + \frac{b_{t-1}^S}{\pi_t} + N_t, \tag{2}$$

where Φ_t, N_t are exogenous. Choice variables: $\{c_t, b_t^S\}_{t\geq 0}$.

2.2 Lagrangian

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{c_t^{1-\sigma}}{1-\sigma} - \mu_t \left[c_t + Q_t^S b_t^S + \Phi_t - y_t + \tau_t - \frac{b_{t-1}^S}{\pi_t} - N_t \right] \right\}. \tag{3}$$

2.3 First-order conditions

w.r.t. c_t: $\partial\mathcal{L}/\partial c_t = 0$:

$$c_t^{-\sigma} - \mu_t = 0 \implies \mu_t = c_t^{-\sigma}. \tag{4}$$

w.r.t. b_t^S: The variable b_t^S appears in the period-t constraint with coefficient $-Q_t^S$ on the LHS (entering as $+Q_t^S b_t^S$ inside the bracket multiplied by $-\mu_t$) and in the period-$(t+1)$ constraint with coefficient $+1/\pi_{t+1}$ (entering as $-b_t^S/\pi_{t+1}$ inside the bracket multiplied by $-\mu_{t+1}$, hence sign flip).

The relevant terms in $\mathcal{L}$ involving b_t^S:

$$-\beta^t \mu_t Q_t^S b_t^S + \beta^{t+1} \mu_{t+1} \frac{b_t^S}{\pi_{t+1}}. \tag{5}$$

Taking $\partial/\partial b_t^S$ and setting to zero, with conditional expectation at t:

$$-\beta^t \mu_t Q_t^S + \beta^{t+1} E_t\left[\frac{\mu_{t+1}}{\pi_{t+1}}\right] = 0. \tag{6}$$

Divide by β^t:

$$\mu_t Q_t^S = \beta E_t\left[\frac{\mu_{t+1}}{\pi_{t+1}}\right]. \tag{7}$$

Substitute (4):

$$c_t^{-\sigma} Q_t^S = \beta E_t\left[\frac{c_{t+1}^{-\sigma}}{\pi_{t+1}}\right]. \tag{8}$$

Define $M_{t+1} \equiv \beta(c_{t+1}/c_t)^{-\sigma}$. Then (8) becomes

$$\boxed{Q_t^S = E_t\left[\frac{M_{t+1}}{\pi_{t+1}}\right].} \tag{9}$$

2.4 Transversality

Standard:

$$\lim_{T\to\infty} E_t\big[\beta^T c_T^{-\sigma} Q_T^S b_T^S\big] = 0. \tag{10}$$

3 Long-Bond Market: Specification

3.1 Geometric perpetuity payoff structure

A long bond issued at t pays nominal coupons:

$$1 \text{ at } t+1, \quad \delta \text{ at } t+2, \quad \delta^2 \text{ at } t+3, \quad \ldots \tag{11}$$

A bond issued at $t-k$ has remaining payoff stream from date t:

$$\delta^k \text{ at } t+1, \quad \delta^{k+1} \text{ at } t+2, \quad \ldots = \delta^k \cdot (\text{payoff of fresh bond}). \tag{12}$$

Therefore one unit of a k-period-old bond is identical to δ^k units of a freshly-issued bond. Define the effective stock B_t^L in fresh-bond units. Law of motion:

$$B_t^L = \delta B_{t-1}^L + B_t^{L,\text{new}}. \tag{13}$$

Single nominal price Q_t^L applies to entire stock.

3.2 Regulatory market-clearing condition

$$\boxed{Q_t^L b_t^L = \lambda_t^L V^{\text{liab}}.} \tag{14}$$

3.3 Counterfactual fundamental price

Define

$$Q_t^{L,\text{fund}} \equiv E_t\left[\frac{M_{t+1}(1+\delta Q_{t+1}^L)}{\pi_{t+1}}\right]. \tag{15}$$

3.4 Wedge

Define ω_t by

$$\boxed{1+\omega_t \equiv \frac{Q_t^L}{Q_t^{L,\text{fund}}}.} \tag{16}$$

Assumption 1 (Binding mandate)**.** *For all $t \geq 0$, $\omega_t > 0$.*

3.5 Per-period rent

Define

$$\mathcal{R}_t \equiv \frac{\omega_t}{1+\omega_t} \cdot Q_t^L b_t^L. \tag{17}$$

Substituting (14):

$$\boxed{\mathcal{R}_t = \frac{\omega_t}{1+\omega_t}\lambda_t^L V^{\text{liab}}.} \tag{18}$$

3.6 Long-end transversality

Asserted:

$$\lim_{T\to\infty} E_t\left[\prod_{k=1}^{T-t} M_{t+k} \cdot Q_T^L b_T^L\right] = 0. \tag{19}$$

4 Resource Constraint

Endowment economy:

$$c_t + g_t = y_t. \tag{20}$$

5 Government

5.1 Nominal flow budget constraint

At date t, nominal obligations $B_{t-1}^S + B_{t-1}^L$ (short-bond principal B_{t-1}^S paid as one nominal unit per bond; long-bond coupon B_{t-1}^L paid as one nominal unit per effective unit). Nominal financing: $P_t s_t$ (real surplus converted to nominal), $Q_t^S B_t^S$ (proceeds from new short bonds), $Q_t^L(B_t^L - \delta B_{t-1}^L)$ (proceeds from new long-bond issuance, using (13)):

$$B_{t-1}^S + B_{t-1}^L = P_t s_t + Q_t^S B_t^S + Q_t^L(B_t^L - \delta B_{t-1}^L). \tag{21}$$

5.2 Real form: derivation

Divide (21) by P_t:

$$\frac{B_{t-1}^S}{P_t} + \frac{B_{t-1}^L}{P_t} = s_t + Q_t^S\frac{B_t^S}{P_t} + Q_t^L\left(\frac{B_t^L}{P_t} - \delta\frac{B_{t-1}^L}{P_t}\right). \tag{22}$$

Use $B_{t-1}^S/P_t = (B_{t-1}^S/P_{t-1}) \cdot (P_{t-1}/P_t) = b_{t-1}^S/\pi_t$, similarly for long bonds, and $B_t^S/P_t = b_t^S$, $B_t^L/P_t = b_t^L$:

$$\frac{b_{t-1}^S}{\pi_t} + \frac{b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L - \delta Q_t^L\frac{b_{t-1}^L}{\pi_t}. \tag{23}$$

Move $-\delta Q_t^L b_{t-1}^L/\pi_t$ to LHS:

$$\frac{b_{t-1}^S}{\pi_t} + \frac{b_{t-1}^L}{\pi_t} + \delta Q_t^L \frac{b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L. \tag{24}$$

Factor:

$$\boxed{\frac{b_{t-1}^S + (1+\delta Q_t^L) b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L.} \tag{25}$$

5.3 Maturity composition

$$b_t^L = \chi_t (b_t^S + b_t^L). \tag{26}$$

6 Augmented FTPL Identity: Derivation

6.1 Definitions

Real market value of debt at start of t:

$$v_t \equiv \frac{b_{t-1}^S + (1+\delta Q_t^L) b_{t-1}^L}{\pi_t}. \tag{27}$$

Real market value at end of t:

$$\tilde{v}_t \equiv Q_t^S b_t^S + Q_t^L b_t^L. \tag{28}$$

By (25):

$$v_t = s_t + \tilde{v}_t. \tag{29}$$

6.2 Step 1: v_{t+1} at one period forward

Apply (27) at $t+1$:

$$v_{t+1} = \frac{b_t^S + (1+\delta Q_{t+1}^L) b_t^L}{\pi_{t+1}}. \tag{30}$$

6.3 Step 2: compute $E_t[M_{t+1} v_{t+1}]$

$$E_t[M_{t+1} v_{t+1}] = E_t\left[\frac{M_{t+1}(b_t^S + (1+\delta Q_{t+1}^L) b_t^L)}{\pi_{t+1}}\right]. \tag{31}$$

Quantities b_t^S, b_t^L are in the date-t information set; pull them out:

$$E_t[M_{t+1} v_{t+1}] = b_t^S \cdot E_t\left[\frac{M_{t+1}}{\pi_{t+1}}\right] + b_t^L \cdot E_t\left[\frac{M_{t+1}(1+\delta Q_{t+1}^L)}{\pi_{t+1}}\right]. \tag{32}$$

Apply (9) to the first term and (15) to the second:

$$E_t[M_{t+1} v_{t+1}] = Q_t^S b_t^S + Q_t^{L,\text{fund}} b_t^L. \tag{33}$$

6.4 Step 3: substitute the wedge

From (16): $Q_t^{L,\text{fund}} = Q_t^L/(1+\omega_t)$. Substitute into (33):

$$E_t[M_{t+1} v_{t+1}] = Q_t^S b_t^S + \frac{Q_t^L b_t^L}{1+\omega_t}. \tag{34}$$

6.5 Step 4: compute $\tilde{v}_t - E_t[M_{t+1}v_{t+1}]$

Subtracting (34) from (28):

$$\tilde{v}_t - E_t[M_{t+1}v_{t+1}] = (Q_t^S b_t^S + Q_t^L b_t^L) - \left(Q_t^S b_t^S + \frac{Q_t^L b_t^L}{1+\omega_t}\right). \tag{35}$$

The $Q_t^S b_t^S$ terms cancel:

$$\tilde{v}_t - E_t[M_{t+1}v_{t+1}] = Q_t^L b_t^L - \frac{Q_t^L b_t^L}{1+\omega_t} = Q_t^L b_t^L \left(1 - \frac{1}{1+\omega_t}\right). \tag{36}$$

Simplify $1 - 1/(1+\omega_t) = \omega_t/(1+\omega_t)$:

$$\tilde{v}_t - E_t[M_{t+1}v_{t+1}] = Q_t^L b_t^L \cdot \frac{\omega_t}{1+\omega_t} = \mathcal{R}_t. \tag{37}$$

6.6 Step 5: combine with flow constraint

From (29): $\tilde{v}_t = v_t - s_t$. Substitute into (37):

$$v_t - s_t - E_t[M_{t+1}v_{t+1}] = \mathcal{R}_t. \tag{38}$$

Rearrange:

$$\boxed{v_t = s_t + \mathcal{R}_t + E_t[M_{t+1}v_{t+1}].} \tag{39}$$

6.7 Step 6: forward iteration

Start from (39). Apply at $t+1$:

$$v_{t+1} = s_{t+1} + \mathcal{R}_{t+1} + E_{t+1}[M_{t+2}v_{t+2}]. \tag{40}$$

Multiply by M_{t+1} and take E_t:

$$\begin{aligned} E_t[M_{t+1}v_{t+1}] &= E_t[M_{t+1}(s_{t+1} + \mathcal{R}_{t+1})] + E_t[M_{t+1}E_{t+1}[M_{t+2}v_{t+2}]] & (41)\\ &= E_t[M_{t+1}(s_{t+1} + \mathcal{R}_{t+1})] + E_t[M_{t+1}M_{t+2}v_{t+2}], & (42)\end{aligned}$$

using law of iterated expectations $E_t[E_{t+1}[\cdot]] = E_t[\cdot]$ together with the fact that M_{t+1} is in the $t+1$-information set.

Substitute back into (39):

$$v_t = s_t + \mathcal{R}_t + E_t[M_{t+1}(s_{t+1} + \mathcal{R}_{t+1})] + E_t[M_{t+1}M_{t+2}v_{t+2}]. \tag{43}$$

Iterate T times. By induction, with the convention $\prod_{k=1}^{0} M_{t+k} = 1$:

$$v_t = E_t \sum_{j=0}^{T} \left(\prod_{k=1}^{j} M_{t+k}\right)(s_{t+j} + \mathcal{R}_{t+j}) + E_t\left[\prod_{k=1}^{T+1} M_{t+k} \cdot v_{t+T+1}\right]. \tag{44}$$

Verification of induction. The claim holds at $T = 0$ trivially: $v_t = s_t + \mathcal{R}_t + E_t[M_{t+1}v_{t+1}]$. Assume it holds at T. Apply (39) at $t + T + 1$:

$$v_{t+T+1} = s_{t+T+1} + \mathcal{R}_{t+T+1} + E_{t+T+1}[M_{t+T+2}v_{t+T+2}]. \tag{45}$$

Multiply by $\prod_{k=1}^{T+1} M_{t+k}$ and take E_t:

$$E_t\left[\prod_{k=1}^{T+1} M_{t+k} v_{t+T+1}\right] = E_t\left[\prod_{k=1}^{T+1} M_{t+k}(s_{t+T+1} + \mathcal{R}_{t+T+1})\right] \tag{46}$$

$$+ E_t\left[\prod_{k=1}^{T+2} M_{t+k} v_{t+T+2}\right]. \tag{47}$$

Substituting into the T-th formula gives the $T+1$-th formula. □

6.8 Step 7: transversality on the residual

We require:

$$\lim_{T\to\infty} E_t\left[\prod_{k=1}^{T+1} M_{t+k} \cdot v_{t+T+1}\right] = 0. \tag{48}$$

Using (27) at $t+T+1$ (with index $T+1$):

$$v_{t+T+1} = \frac{b^S_{t+T} + (1+\delta Q^L_{t+T+1}) b^L_{t+T}}{\pi_{t+T+1}}. \tag{49}$$

Then

$$\prod_{k=1}^{T+1} M_{t+k} \cdot v_{t+T+1} = \prod_{k=1}^{T+1} M_{t+k} \cdot \frac{b^S_{t+T} + (1+\delta Q^L_{t+T+1}) b^L_{t+T}}{\pi_{t+T+1}}. \tag{50}$$

Recall M_{t+k}/π_{t+k} is the nominal SDF for one period. The product $\prod_{k=1}^{T+1} M_{t+k}/\pi_{t+T+1}$ is a real-to-real discount factor multiplied by an extra $1/\pi_{t+T+1}$ which converts the start-of-$(t+T+1)$ nominal payoff to a real payoff at $t+T+1$ prices.

To reduce to standard transversality conditions, use the date-$(t+T)$ price relations. By (9): $Q^S_{t+T} = E_{t+T}[M_{t+T+1}/\pi_{t+T+1}]$. By (15): $Q^{L,\text{fund}}_{t+T} = E_{t+T}[M_{t+T+1}(1+\delta Q^L_{t+T+1})/\pi_{t+T+1}]$.

Take E_t of $\prod_{k=1}^{T+1} M_{t+k} \cdot v_{t+T+1}$. By the law of iterated expectations through date $t+T$:

$$E_t\left[\prod_{k=1}^{T+1} M_{t+k} v_{t+T+1}\right] = E_t\left[\prod_{k=1}^{T} M_{t+k} \cdot E_{t+T}[M_{t+T+1} v_{t+T+1}]\right]. \tag{51}$$

Compute the inner expectation:

$$E_{t+T}[M_{t+T+1} v_{t+T+1}] = E_{t+T}\left[\frac{M_{t+T+1}(b^S_{t+T} + (1+\delta Q^L_{t+T+1}) b^L_{t+T})}{\pi_{t+T+1}}\right] \tag{52}$$

$$= b^S_{t+T} E_{t+T}\left[\frac{M_{t+T+1}}{\pi_{t+T+1}}\right] + b^L_{t+T} E_{t+T}\left[\frac{M_{t+T+1}(1+\delta Q^L_{t+T+1})}{\pi_{t+T+1}}\right] \tag{53}$$

$$= Q^S_{t+T} b^S_{t+T} + Q^{L,\text{fund}}_{t+T} b^L_{t+T}. \tag{54}$$

Use $Q^{L,\text{fund}}_{t+T} = Q^L_{t+T}/(1+\omega_{t+T})$:

$$E_{t+T}[M_{t+T+1} v_{t+T+1}] = Q^S_{t+T} b^S_{t+T} + \frac{Q^L_{t+T} b^L_{t+T}}{1+\omega_{t+T}}. \tag{55}$$

Therefore:

$$E_t\left[\prod_{k=1}^{T+1} M_{t+k} v_{t+T+1}\right] = E_t\left[\prod_{k=1}^{T} M_{t+k}\left(Q^S_{t+T} b^S_{t+T} + \frac{Q^L_{t+T} b^L_{t+T}}{1+\omega_{t+T}}\right)\right]. \tag{56}$$

The two terms in the bracket are non-negative (since $Q^S, Q^L, b^S, b^L > 0$ and $\omega > 0$, hence $1/(1+\omega) \in (0,1)$). Therefore (48) is implied by:

$$\lim_{T\to\infty} E_t\left[\prod_{k=1}^{T} M_{t+k} \cdot Q^S_{t+T} b^S_{t+T}\right] = 0, \tag{57}$$

$$\lim_{T\to\infty} E_t\left[\prod_{k=1}^{T} M_{t+k} \cdot \frac{Q^L_{t+T} b^L_{t+T}}{1+\omega_{t+T}}\right] = 0. \tag{58}$$

(57) follows from the household transversality (10) after rewriting $\beta^T c_T^{-\sigma}$ in terms of products of M_{t+k}:

$$\beta^T c_T^{-\sigma} = c_t^{-\sigma} \cdot \prod_{k=1}^{T} \beta(c_{t+k}/c_{t+k-1})^{-\sigma} = c_t^{-\sigma} \prod_{k=1}^{T} M_{t+k}, \tag{59}$$

provided we set the indexing so that T in (10) corresponds to date $t+T$.[1]

(58) follows from (19) together with the bound $1/(1+\omega_{t+T}) < 1$ (which makes (58) a smaller-magnitude limit than (19), hence implied).

6.9 Step 8: take the limit

Taking $T \to \infty$ in (44) and using (48):

$$\boxed{v_t = E_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^{j} M_{t+k}\right)(s_{t+j} + \mathcal{R}_{t+j}).} \tag{60}$$

7 Steady State

Set $\bar{\pi} = 1$, drop time subscripts, all variables constant. Output $\bar{y}$ exogenous, so $c = \bar{y} - g$, $M = \beta(c/c)^{-\sigma} = \beta$.

7.1 Short-bond price

From (9) with $M = \beta$ and $\bar{\pi} = 1$:

$$Q^S = \beta. \tag{61}$$

7.2 Long-bond price from market clearing

From (14):

$$Q^L = \frac{\lambda^L V^{\text{liab}}}{b^L}. \tag{62}$$

[1]The factor $c_t^{-\sigma}$ is finite and constant from the perspective of date t, so multiplying or dividing the limit by it does not affect whether it is zero.

7.3 Long-bond price from no-arbitrage with wedge

From (15) with $M = \beta$, $\bar{\pi} = 1$, $Q^L_{t+1} = Q^L$:

$$Q^{L,\text{fund}} = \beta(1 + \delta Q^L). \tag{63}$$

From (16): $Q^L = (1+\omega)Q^{L,\text{fund}}$. Combine:

$$Q^L = (1+\omega)\beta(1 + \delta Q^L). \tag{64}$$

7.4 Solving for ω in closed form

Define $z \equiv (1+\omega)\beta$. Equation (64):

$$Q^L = z(1 + \delta Q^L) = z + z\delta Q^L. \tag{65}$$

Move $z\delta Q^L$ to LHS:

$$Q^L - z\delta Q^L = z \implies Q^L(1 - z\delta) = z \implies Q^L = \frac{z}{1 - z\delta}, \tag{66}$$

valid provided $z\delta < 1$.

Equate (62) and (66). Define $X \equiv \lambda^L V^{\text{liab}}/b^L$:

$$\frac{z}{1 - z\delta} = X. \tag{67}$$

Cross-multiply:

$$z = X(1 - z\delta) = X - Xz\delta. \tag{68}$$

Move $Xz\delta$ to LHS:

$$z + Xz\delta = X \implies z(1 + X\delta) = X \implies z = \frac{X}{1 + X\delta}. \tag{69}$$

Substitute back $z = (1+\omega)\beta$:

$$(1+\omega)\beta = \frac{X}{1 + X\delta} \implies 1 + \omega = \frac{X}{\beta(1 + X\delta)}. \tag{70}$$

Subtract one:

$$\omega = \frac{X}{\beta(1 + X\delta)} - 1 = \frac{X - \beta(1 + X\delta)}{\beta(1 + X\delta)} = \frac{X - \beta - X\beta\delta}{\beta(1 + X\delta)} = \frac{X(1 - \beta\delta) - \beta}{\beta(1 + X\delta)}. \tag{71}$$

Substitute back $X = \lambda^L V^{\text{liab}}/b^L$. Numerator:

$$X(1 - \beta\delta) - \beta = \frac{\lambda^L V^{\text{liab}}(1 - \beta\delta)}{b^L} - \beta = \frac{\lambda^L V^{\text{liab}}(1 - \beta\delta) - \beta b^L}{b^L}. \tag{72}$$

Denominator:

$$\beta(1 + X\delta) = \beta\left(1 + \frac{\delta\lambda^L V^{\text{liab}}}{b^L}\right) = \frac{\beta(b^L + \delta\lambda^L V^{\text{liab}})}{b^L}. \tag{73}$$

Take the ratio (the b^L in numerator and denominator cancel):

$$\boxed{\omega = \frac{\lambda^L V^{\text{liab}}(1 - \beta\delta) - \beta b^L}{\beta(b^L + \delta\lambda^L V^{\text{liab}})}.} \tag{74}$$

7.5 Closed-form rent

From (18):

$$\mathcal{R} = \frac{\omega}{1+\omega}\lambda^L V^{\text{liab}}. \tag{75}$$

Compute $\omega/(1+\omega)$. From (70): $1+\omega = X/[\beta(1+X\delta)]$. Therefore $1/(1+\omega) = \beta(1+X\delta)/X$. So:

$$\frac{\omega}{1+\omega} = 1 - \frac{1}{1+\omega} = 1 - \frac{\beta(1+X\delta)}{X} = \frac{X-\beta(1+X\delta)}{X} = \frac{X(1-\beta\delta)-\beta}{X}. \tag{76}$$

Multiply by $\lambda^L V^{\text{liab}}$. Use $X = \lambda^L V^{\text{liab}}/b^L$, so $\lambda^L V^{\text{liab}} = Xb^L$:

$$\mathcal{R} = \frac{X(1-\beta\delta)-\beta}{X} \cdot Xb^L = [X(1-\beta\delta)-\beta]b^L. \tag{77}$$

Substitute $X = \lambda^L V^{\text{liab}}/b^L$:

$$\mathcal{R} = \left[\frac{\lambda^L V^{\text{liab}}(1-\beta\delta)}{b^L} - \beta\right] b^L = \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L. \tag{78}$$

$$\boxed{\mathcal{R} = \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L.} \tag{79}$$

This is exactly the numerator of ω in (74).

7.6 Steady-state augmented FTPL identity

In steady state with $\bar{\pi} = 1$, equation (60) reduces to a geometric sum. Constants $s, \mathcal{R}$, and $\prod_{k=1}^{j} M_{t+k} = \beta^j$:

$$v = (s+\mathcal{R})\sum_{j=0}^{\infty}\beta^j = \frac{s+\mathcal{R}}{1-\beta}. \tag{80}$$

Compute v from (27) with $\bar{\pi} = 1$:

$$v = b^S_{t-1} + (1+\delta Q^L)b^L_{t-1} = b^S + (1+\delta Q^L)b^L = b^S + b^L + \delta Q^L b^L, \tag{81}$$

where the second equality uses constancy. Substitute $Q^L b^L = \lambda^L V^{\text{liab}}$:

$$v = b^S + b^L + \delta\lambda^L V^{\text{liab}}. \tag{82}$$

Combining (80) and (82):

$$\boxed{b^S + b^L + \delta\lambda^L V^{\text{liab}} = \frac{s + \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L}{1-\beta}.} \tag{83}$$

7.7 Long-end repression share

$$\zeta \equiv \frac{\mathcal{R}/(1-\beta)}{v} = \frac{\lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L}{(1-\beta)(b^S + b^L + \delta\lambda^L V^{\text{liab}})}. \tag{84}$$

8 Comparative Statics in Steady State

Differentiate (74) with respect to parameters. Write $N \equiv \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L$, $D \equiv \beta(b^L + \delta\lambda^L V^{\text{liab}})$, so $\omega = N/D$.

8.1 $\partial\omega/\partial\lambda^L$

$$\frac{\partial N}{\partial\lambda^L} = V^{\text{liab}}(1-\beta\delta), \tag{85}$$
$$\frac{\partial D}{\partial\lambda^L} = \beta\delta V^{\text{liab}}. \tag{86}$$

Quotient rule:

$$\frac{\partial\omega}{\partial\lambda^L} = \frac{(\partial N/\partial\lambda^L)D - N(\partial D/\partial\lambda^L)}{D^2}. \tag{87}$$

Numerator:

$$V^{\text{liab}}(1-\beta\delta)\cdot\beta(b^L+\delta\lambda^L V^{\text{liab}}) - [\lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L]\cdot\beta\delta V^{\text{liab}} \tag{88}$$
$$= \beta V^{\text{liab}}\left\{(1-\beta\delta)(b^L+\delta\lambda^L V^{\text{liab}}) - \delta[\lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L]\right\}. \tag{89}$$

Expand inside the braces:

$$(1-\beta\delta)b^L + (1-\beta\delta)\delta\lambda^L V^{\text{liab}} - \delta\lambda^L V^{\text{liab}}(1-\beta\delta) + \delta\beta b^L \tag{90}$$
$$= (1-\beta\delta)b^L + \delta\beta b^L \tag{91}$$
$$= b^L[(1-\beta\delta)+\beta\delta] \tag{92}$$
$$= b^L. \tag{93}$$

Therefore:

$$\frac{\partial\omega}{\partial\lambda^L} = \frac{\beta V^{\text{liab}}\cdot b^L}{D^2} = \frac{V^{\text{liab}} b^L}{\beta(b^L+\delta\lambda^L V^{\text{liab}})^2} > 0. \tag{94}$$

8.2 $\partial\omega/\partial b^L$

$$\frac{\partial N}{\partial b^L} = -\beta, \tag{95}$$
$$\frac{\partial D}{\partial b^L} = \beta. \tag{96}$$

$$\frac{\partial\omega}{\partial b^L} = \frac{(-\beta)D - N\cdot\beta}{D^2} = \frac{-\beta(D+N)}{D^2}. \tag{97}$$

Compute $D+N$:

$$D+N = \beta(b^L+\delta\lambda^L V^{\text{liab}}) + \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L \tag{98}$$
$$= \beta b^L + \beta\delta\lambda^L V^{\text{liab}} + \lambda^L V^{\text{liab}} - \beta\delta\lambda^L V^{\text{liab}} - \beta b^L \tag{99}$$
$$= \lambda^L V^{\text{liab}}. \tag{100}$$

Therefore:

$$\frac{\partial\omega}{\partial b^L} = \frac{-\beta\lambda^L V^{\text{liab}}}{D^2} = \frac{-\lambda^L V^{\text{liab}}}{\beta(b^L+\delta\lambda^L V^{\text{liab}})^2} < 0. \tag{101}$$

8.3 $\partial\omega/\partial V^{\text{liab}}$

By symmetry of λ^L and V^{liab} in N and D (both enter only via the product $\lambda^L V^{\text{liab}}$), the calculation is analogous to (94):

$$\frac{\partial\omega}{\partial V^{\text{liab}}} = \frac{\lambda^L b^L}{\beta(b^L + \delta\lambda^L V^{\text{liab}})^2} > 0. \tag{102}$$

8.4 $\partial\omega/\partial\chi$ holding total debt b fixed

Set $b^L = \chi b$ with b fixed. By the chain rule:

$$\left.\frac{\partial\omega}{\partial\chi}\right|_b = \frac{\partial\omega}{\partial b^L}\cdot b = \frac{-\lambda^L V^{\text{liab}} b}{\beta(\chi b + \delta\lambda^L V^{\text{liab}})^2} < 0. \tag{103}$$

8.5 $\partial\mathcal{R}/\partial\chi$ holding total debt b fixed

From (79): $\mathcal{R} = \lambda^L V^{\text{liab}}(1-\beta\delta) - \beta b^L$. With $b^L = \chi b$:

$$\left.\frac{\partial\mathcal{R}}{\partial\chi}\right|_b = -\beta b < 0. \tag{104}$$

Alternatively, from $\mathcal{R} = [\omega/(1+\omega)]\lambda^L V^{\text{liab}}$, differentiating using $\partial[\omega/(1+\omega)]/\partial\chi = (1+\omega)^{-2}\partial\omega/\partial\chi$:

$$\left.\frac{\partial\mathcal{R}}{\partial\chi}\right|_b = \frac{\lambda^L V^{\text{liab}}}{(1+\omega)^2}\cdot\frac{\partial\omega}{\partial\chi} < 0, \tag{105}$$

consistent with (104).

9 Summary System

Endogenous variables ($t \geq 0$): $c_t, b_t^S, b_t^L, Q_t^S, Q_t^L, Q_t^{L,\text{fund}}, \omega_t, \pi_t, P_t, \mathcal{R}_t, v_t$.

Exogenous ($t \geq 0$): $y_t, g_t, \tau_t, s_t, \lambda_t^L, \chi_t, b_t$, and structural constants $V^{\text{liab}}, \beta, \sigma, \delta$.

Initial conditions: $b_{-1}^S, b_{-1}^L, P_{-1}$.

Equations (counted to match endogenous variables):

$$c_t + g_t = y_t \tag{R}$$

$$M_{t+1} = \beta(c_{t+1}/c_t)^{-\sigma} \tag{def}$$

$$Q_t^S = E_t[M_{t+1}/\pi_{t+1}] \tag{H2}$$

$$Q_t^L b_t^L = \lambda_t^L V^{\text{liab}} \tag{LDI-MC}$$

$$Q_t^{L,\text{fund}} = E_t[(M_{t+1}/\pi_{t+1})(1+\delta Q_{t+1}^L)] \tag{LDI-F}$$

$$1+\omega_t = Q_t^L/Q_t^{L,\text{fund}} \tag{LDI-W}$$

$$\mathcal{R}_t = [\omega_t/(1+\omega_t)]\lambda_t^L V^{\text{liab}} \tag{RENT}$$

$$\frac{b_{t-1}^S + (1+\delta Q_t^L)b_{t-1}^L}{\pi_t} = s_t + Q_t^S b_t^S + Q_t^L b_t^L \tag{G1}$$

$$b_t^L = \chi_t(b_t^S + b_t^L) \tag{G2}$$

$$v_t = \frac{b_{t-1}^S + (1+\delta Q_t^L)b_{t-1}^L}{\pi_t} \tag{def}$$

$$P_t = \pi_t P_{t-1} \tag{def}$$

Implicit constraints:

$$\lim_{T\to\infty} E_t[\beta^T c_T^{-\sigma} Q_T^S b_T^S] = 0 \tag{H3}$$

$$\lim_{T\to\infty} E_t[(\prod_{k=1}^{T-t} M_{t+k}) Q_T^L b_T^L] = 0 \tag{LDI-TVC}$$

Implied identity:

$$v_t = E_t \sum_{j=0}^{\infty} \left(\prod_{k=1}^{j} M_{t+k} \right) (s_{t+j} + \mathcal{R}_{t+j}). \tag{FTPL}$$